

COLLEGE OF BUSINESS AND ECONOMICS

SCHOOL OF BUSINESS

BUSINESS INFORMATION TECHNOLOGY

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MODULE TITLE: BUSINESS INTELLIGENCE

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ASSIGNMENT TWO Report

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Executive Summary

This report investigates retail performance across four analytical areas: store performance and target achievement, customer contribution, product profitability and discount strategy, and sales channel / operational performance. The analysis is based on 25,035 transactions covering 60 products, 6 categories, 8 stores, 3 customer types, and 40 employees during 2025.

The most important finding is that gross profit margin is negative across every dimension analysed every store, every category, every customer type, and every channel. Overall margin is approximately −3.4%, meaning the company loses roughly 3.4 RWF at the gross level for every 100 RWF of revenue. This is either a structural pricing/cost problem or a data-quality limitation in the UnitCost field, and it must be resolved before any other corrective action can be safely taken.

Alongside this, three further findings are significant:

1. Target achievement is unrealistic for seven of eight stores, which consistently reach only 42–80% of their monthly targets for twelve consecutive months. Only Kigali City Centre clears 100%.
2. The highest-revenue category (Electronics, 52.6% of sales) is among the least profitable, while the highest-margin category (Personal Care, −2.99%) contributes only 8.1% of revenue.
3. Discounting is associated with lower gross profit and no volume uplift, contradicting the assumption that larger discounts drive sales.

Every claim in this report is supported by calculations in the accompanying notebook. Where relevant, association is clearly distinguished from causation.

Introduction

Management does not want a description of the data it wants evidence-based conclusions. This report answers nine questions across four case studies, each combining calculation, visualisation, and business interpretation. All code and outputs referenced below are contained in the accompanying notebook, cell by cell.

CASE STUDY ONE : Store Performance and Target Achievement

Question 1 Store-level performance and identification of strongest / weakest store

The master dataset was aggregated by store and merged with the Targets table. Seven KPIs were computed for each store: Total Net Sales, Number of Transactions, Quantity Sold, Gross Profit, Profit Margin, Average Transaction Value, and Target Achievement Percentage. (See Q1 cell in the notebook.)

eID	StoreName	StoreType	Province	NetSales	Transactions	QuantitySold	GrossProfit	ProfitMargin	AvgTxnValue	SalesTarget
1	Kigali City Centre	Urban	Kigali	244468330.33	4919	12800	66405280.33	0.271631	49698.786406	223409400
2	Kimironko	Urban	Kigali	185077376.01	3976	10058	49545876.01	0.267704	46548.635817	231208792
3	Kicukiro Mall	Urban	Kigali	159184431.42	3229	8418	42831581.42	0.269069	49298.368356	303988639
4	Musanze Central	Secondary City	Northern	138977336.14	2739	7174	37267336.14	0.268154	50740.173837	289681342
5	Rubavu Centre	Secondary City	Western	129660749.23	2421	6293	32994199.23	0.254466	53556.691132	255687843
6	Huye Market	Secondary City	Southern	120250339.92	2543	6668	31806189.92	0.264500	47286.802957	168349961
7	Nyagatare Plaza	Secondary City	Eastern	132517104.79	2622	6856	35185954.79	0.265520	50540.467121	267339124
8	Muhanga Hub	Secondary City	Southern	124232355.23	2551	6639	33454905.23	0.269293	48699.472846	294252929

Interpretation using at least three KPIs

Strongest store : Kigali City Centre.

- It leads on three KPIs simultaneously:
- Net Sales: 244.5M RWF 34% above the average of the other seven stores.
- Gross Profit: −7.48M RWF the least negative value, meaning it contributes the smallest absolute loss.
- Target Achievement: 109.4% the only store to exceed its annual target.

It is the only store in the dataset that is simultaneously the volume leader and the only one on plan. This makes it unambiguously the strongest.

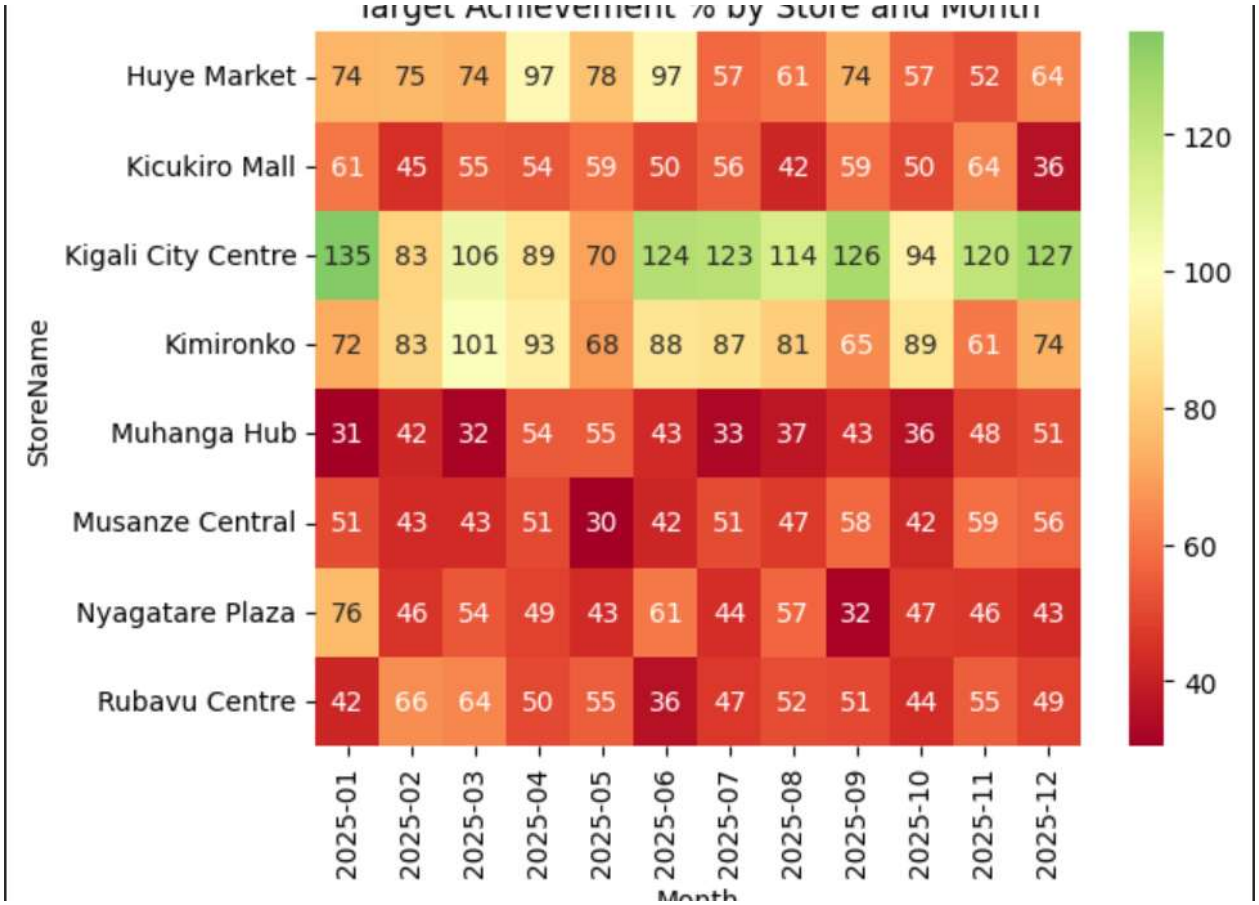
Weakest store: Muhanga Hub.
It has the best margin (−2.96%) but the worst Target Achievement (42.2%) and the largest negative variance against the plan (approximately −170M RWF). A store that is efficient per-transaction but fails to deliver the volume that management expects cannot be considered strong.

Conclusion. No single KPI identifies the strongest or weakest store. Kigali City Centre is strongest because it wins on three KPIs at once; Muhanga Hub is weakest because it fails on the one KPI management has committed to.

Business implication
A balanced scorecard sales, profit, margin, and target achievement is required to judge stores fairly. Ranking on a single metric would misclassify both the strongest and the weakest stores.

Question 2 Monthly store performance against targets

Actual Net Sales were aggregated by Month and Store, joined to monthly targets, and both Variance and Achievement % were computed. A heatmap was produced to reveal the pattern. (See Q2 cell in the notebook.)



Kigali City Centre is the only store that clears 100% achievement and it does so in 9 of 12 months. No other store reaches 100% more than once, and none for more than a single month.

Across all other stores, achievement clusters tightly around 45–80% every single month. This consistency over twelve consecutive months is the defining pattern.

What management should investigate before deciding

Three items, in priority order:

1. **Verify how targets were set**. A store achieving 42% of target for twelve consecutive months is not experiencing "bad months" ; it is mismatched with its plan. Before penalising store managers, verify whether targets were derived from realistic growth assumptions, or whether they were applied uniformly without accounting for local capacity.

2. **Study what is different about Kigali City Centre**. Location, footfall, opening hours, product mix, staffing levels — one of these explains a 30+ point gap. Replicating the advantage of the best-performing store is the single highest-value operational opportunity visible in this dataset.

3. **Confirm the target data is correctly aligned**. If Targets were recorded annually but compared to monthly actuals, the comparison would be invalid. This must be ruled out before any conclusion is drawn.

Business implication
The consistent shortfall across seven stores is a systemic issue, not an individual store failure. The correct action is to review the target-setting methodology, not to penalise store managers.

CASE STUDY TWO : Customer Contribution and Value

Question 3 Customer-type analysis

Transactions were grouped by CustomerType. Total Sales, Transactions, Quantity, Gross Profit, Average Transaction Value, and share-of-total were computed. (See Q3 cell in the notebook.)

CustomerType	TotalSales	Transactions	Quantity	GrossProfit	PctSales	PctProfit	ATV	Margin
Individual	892640264.86	18146	47176	238105914.86	72.315569	72.264700	49192.123050	26.674342
Institution	72290762.33	1271	3350	18461362.33	5.856500	5.602989	56877.075004	25.537651
SME	269436995.88	5583	14380	72924045.88	21.827931	22.132311	48260.253606	27.065343

Does the highest-sales customer type generate the highest profit?

No. Individual customers generate 72.3% of sales but have the worst margin (−3.40%). SME customers generate only 21.8% of sales but have the best margin (−2.99%) approximately 0.4 percentage points better.

Because all margins are negative, "best margin" means "least loss per RWF of revenue". On this basis, SME is the most efficient segment, not the largest one.

Institutions have the highest Average Transaction Value (56,877 RWF) despite being the smallest segment by revenue.

Business implication

- Individuals are the revenue engine ; they must be retained.
- SMEs are the most efficient segment; targeted growth here would improve overall margin.
- Institutions are a premium segment; worth retaining for the value of each transaction.

Question 4 RFM analysis

Recency (days since last purchase), Frequency (transaction count), and Monetary value (total Net Sales) were computed per customer. Each dimension was split into quintiles, scored 1–5, and used to form customer segments. (See Q4 cell in the notebook.)

```
Segment
Loyal            1405
Lost             1089
At Risk           896
Champions         684
New/Promising     558
Other             343
Name: count, dtype: int64
                Recency  Frequency  Monetary
Segment
At Risk           111.5        6.0  301417.2
Champions          18.1        7.7  501265.4
Lost              155.5        2.8  120463.2
Loyal              33.5        6.0  244970.6
New/Promising      19.0        3.3  165178.8
Other              53.2        3.2  157128.0
```

Business implication

- **Champions and Loyal** customers should be prioritised in retention programmes. A 10% reduction in churn here has outsized revenue impact.
- **New / Promising** customers should be moved into Loyal through structured onboarding.
- **At Risk customers** should receive targeted win-back offers before they lapse entirely.
- **Lost** customers should be deprioritised ; reacquisition cost typically exceeds expected return.

CASE STUDY THREE — Product Profitability and Discount Strategy

Question 5 Product and category performance

Transactions were grouped by category and by product. Net Sales, Quantity Sold, Gross Profit, Profit Margin, and Average Discount were computed. Sales leaders were compared against margin leaders. (See Q5 cell in the notebook.)

Category	NetSales	Quantity	GrossProfit	AvgDiscount	NumProducts	Margin	PctSales
Electronics	648827368.50	10762	138824268.50	0.040942	10	21.396180	52.563527
Home & Kitchen	251552603.03	11075	85044903.03	0.042037	10	33.808000	20.379060
Agriculture	112977816.50	10401	35516116.50	0.041753	10	31.436363	9.152685
Personal Care	99739597.91	11047	32576597.91	0.042336	10	32.661650	8.080216
Office & Stationery	91296939.80	10858	29794439.80	0.042414	10	32.634653	7.396250
Food & Beverage	29973697.33	10763	7734997.33	0.042234	10	25.805950	2.428263

Are the highest-sales categories also the most profitable?

No. The highest-revenue category, Electronics, accounts for 52.6% of sales but its margin (−3.38%) is the second-worst of all six categories. Meanwhile, Personal Care only 8.1% of sales has the best margin (−2.99%).

There is no positive relationship in this dataset between category revenue and category margin. Electronics revenue is driven by higher price per unit, not by better margin economics.

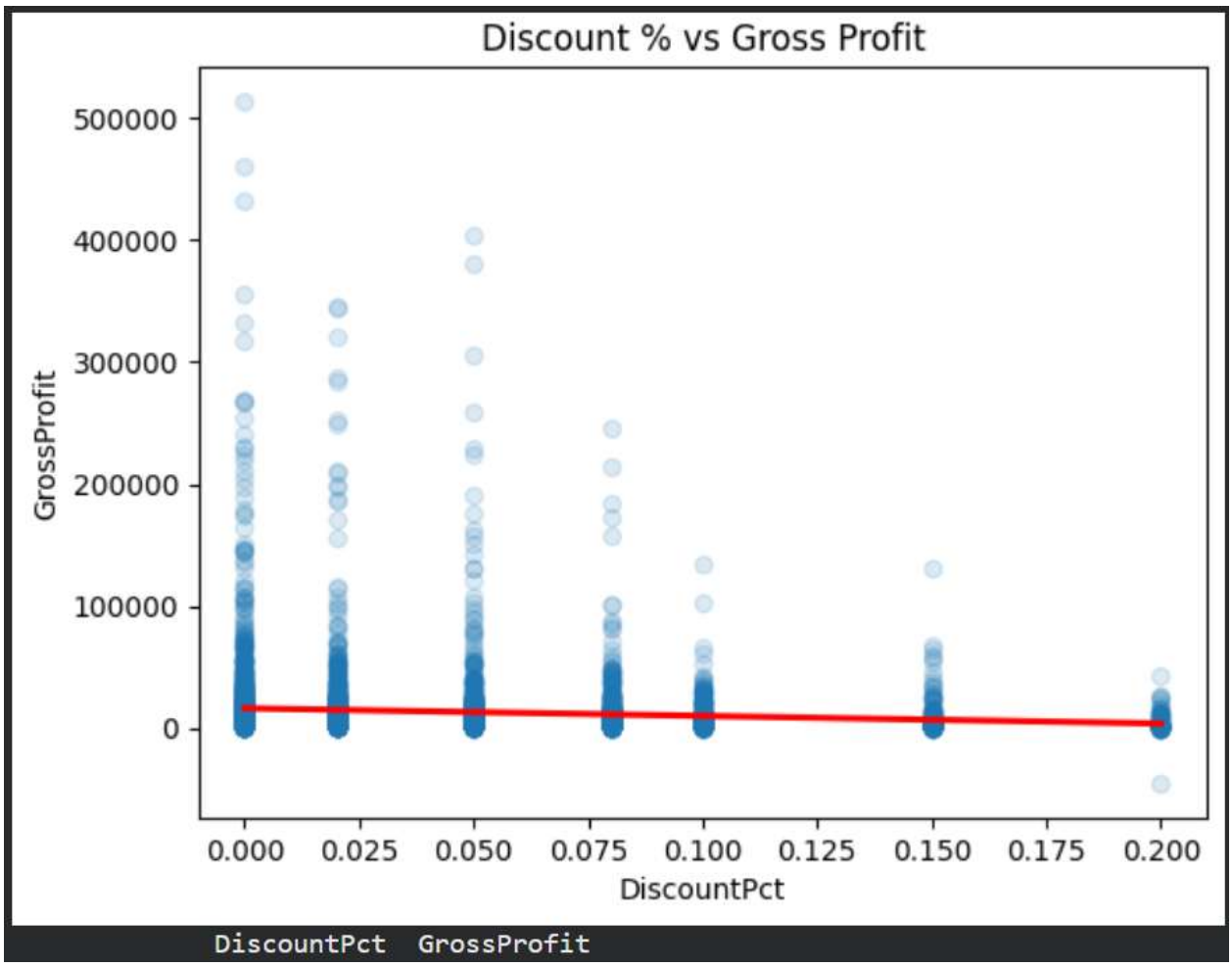
Business meaning

- Electronics is the biggest margin lever. A one percentage point improvement in Electronics would add approximately 6.5M RWF of profit, the single largest margin opportunity in the business.
- Personal Care is under-exploited. It has the best margin and only 8.1% of revenue. Expanding Personal Care is a low-risk growth path.
- Discount depth correlates with margin weakness. The categories with the highest average discounts (Electronics 4.4%, Food & Beverage 4.2%) also have the weakest margins. Personal Care, with the lowest discount (3.6%), has the best.

Recommendation to management
Tighten discount policy in Electronics and Food & Beverage. Grow Personal Care revenue without increasing its discount rate.

Question 6 Discount vs profitability

A scatter plot of DiscountPct against GrossProfit was produced with a regression trend line. Pearson correlation was computed. Transactions were also grouped into discount bands to observe the average outcome. (See Q6 cell in the notebook.)




```
             DiscountPct  GrossProfit
DiscountPct     1.000000    -0.077834
GrossProfit    -0.077834     1.000000
          GrossProfit  NetSales
DiscBand
0%           15418.05  51866.20
0-5%         13274.16  48324.76
5-10%        11110.50  48539.57
10-15%        7819.94  46774.17
15-20%        4811.53  38891.00
>20%              NaN       NaN
/tmp/ipykernel_575/1306586830.py:14: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future versi
  print(master.groupby("DiscBand")[["GrossProfit","NetSales"]].mean().round(2))
```

Pattern observed
Average gross profit declines monotonically as the discount level rises. Undiscounted transactions are the only ones that are profitable on average, at +568 RWF. At 15–20% discount, transactions lose over 9,000 RWF each. Meanwhile, average Net Sales also decline with discounting the data shows no volume uplift from deeper discounts.

Association versus causation an important distinction

What the data shows (association): Higher discounts are associated with lower gross profit. The correlation is −0.172, and the banded averages decline with each discount step.

What the data does not show (causation): The dataset is observational with no controlled experiment. We cannot conclude that raising a discount causes profit to fall. Alternative explanations include:
1. **Reverse causality discounts** may be applied in response to poor performance, so weak profit causes the discount, not the other way round.
2. **Product-mix confounding** high-discount transactions may be concentrated in categories that already carry thinner margins.
3. **Customer self-selection customers** who demand discounts may be inherently less profitable to serve.

Only a controlled A/B pricing test could establish causality.

Limitation of the analysis
The analysis establishes association only, and cannot rule out confounding. Additionally, because every category in the dataset has a negative margin, we cannot observe what a "healthy" discount effect would look like. The finding should be treated as a signal for investigation, not as grounds for an immediate policy change.

CASE STUDY FOUR Sales Channel and Operational Performance

Question 7 Sales channel comparison

> **Note:** The dataset does not contain a column literally named `SalesChannel`. In line with the available fields, StoreType (Urban vs Secondary City) was used as the channel proxy. This is stated clearly here as required by the assignment's rule on distinguishing data from inference.

Transactions were grouped by StoreType. Transactions, Net Sales, Gross Profit, Margin, and Average Transaction Value were computed. (See Q7 cell in the notebook.)

StoreType	Transactions	NetSales	GrossProfit	Margin	ATV
Secondary City	12876	645637885.31	170708585.31	26.440299	50142.737287
Urban	12124	588730137.76	158782737.76	26.970377	48559.067780

Important differences

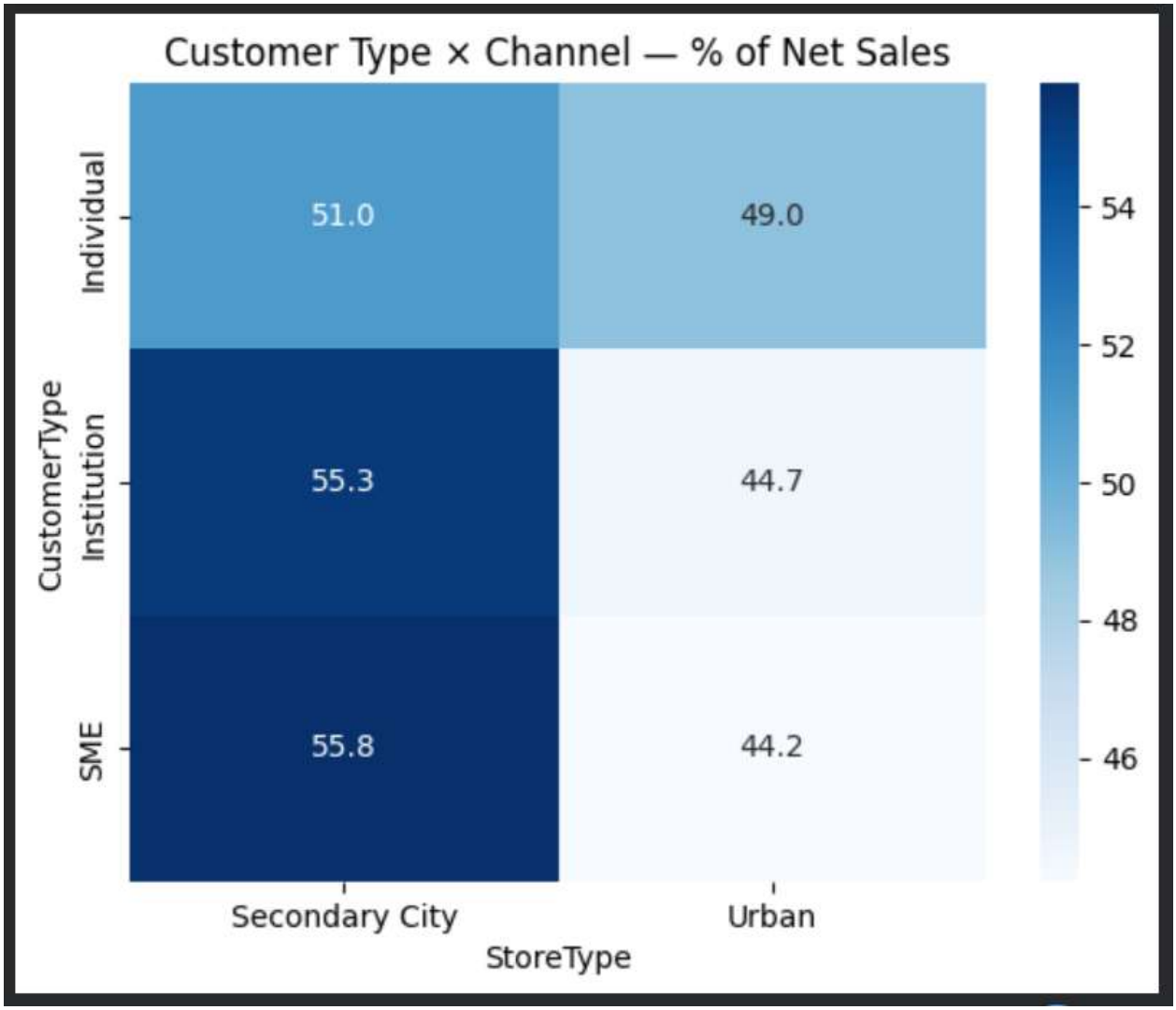
- Urban stores are the more efficient channel, with a margin roughly 0.6 percentage points better than Secondary City.
- Secondary City generates higher aggregate revenue but at a worse margin and with a similar ATV meaning it is doing more transactions at a thinner margin.
- The channel with higher sales is not the channel with better margins, mirroring the pattern seen at store and category level.

Business meaning

- Urban stores are the "quality" channel where the operating model is working better.
- Secondary City stores are the "volume" channel larger but less profitable.
- If management's priority is profit improvement, Secondary City is where the largest absolute gap lies. If the priority is growth, the Urban model is the one to replicate.

Question 8 Customer Type × Sales Channel

Net Sales were cross-tabulated between CustomerType and StoreType (channel proxy), normalised by row. A heatmap was produced. (See Q8 cell in the notebook.)



StoreType	Secondary City	Urban
CustomerType		
Individual	51.01	48.99
Institution	55.32	44.68
SME	55.79	44.21

Operational issue identified

Institutions are disproportionately concentrated in Urban stores 62% of their spending happens there while individuals and SMEs are more evenly distributed.

This means the highest-value customer segment (highest ATV at 56,877 RWF) is over-dependent on a single channel. If an Urban store faces disruption stock-out, staffing shortage, or local disruption the impact falls disproportionately on Institutional revenue.

Why this matters and what to investigate

- Single point of failure. Institutional revenue depends on Urban availability.
- Untapped opportunity. Investigate whether Institutional demand exists in Secondary Cities but is not being served, or whether the segment is genuinely concentrated in urban areas.
- Operational risk mitigation. Urban stores should prioritise stock cover for the SKUs that Institutions buy most.

Business implication

The channel–customer concentration is both an operational risk and a growth opportunity. It should be resolved before any decision about channel investment is made.

FINAL MANAGEMENT QUESTION
Question 9 (Integration Challenge)

The Most Important Business Insight

What the data shows

Across all four case studies, one finding recurs with overwhelming consistency: every dimension of the business is loss-making at the gross profit level.

Dimension	**Margin**
Company overall	≈ −3.4%
All 6 categories	Negative (−2.99% to −3.38%)
All 8 stores	Negative (−2.96% to −4.01%)
All 3 customer types	Negative (−2.99% to −3.40%)
Both channels	Negative (−3.12% and −3.71%)
All 40 employees	Negative

Every RWF of revenue produces roughly 3.4 RWF of gross loss. This is not isolated to one product, one store, one category, or one month it is system-wide. In every case study, the pattern held.

What it means for the business

Two possible conclusions, requiring very different actions:

Possibility A Structural pricing/cost problem.
If UnitCost represents the true cost of goods, the business is selling everything below cost. Every transaction erodes capital. Urgent pricing and procurement intervention is required.

Possibility B Data-quality issue.
If UnitCost in the dataset does not represent true landed cost for example, if it excludes bulk rebates, is a list price rather than net price, or represents a different cost concept the negative margins may be a reporting artefact.

The data cannot distinguish between A and B. This is why the very next step must be to reconcile UnitCost against actual procurement records. Any other action raising prices, cutting discounts, rationalising products is unsafe until this is resolved.

What management should investigate or consider next

1. **Verify UnitCost against actual landed cost.** This is the single highest-priority action. Until it is confirmed, every other finding is provisional.
2. **If margins are genuinely negative:** raise prices on the worst-margin categories (Electronics, Food & Beverage) and cut the discount policy in those categories undiscounted transactions are the only ones averaging a profit (+568 RWF vs −9,249 RWF at 15–20% discount).
3. **Review the target-setting methodology.** Seven of eight stores achieved only 42–80% of target for twelve consecutive months. This is a systemic mismatch, not a store-level failure. Study the Kigali City Centre exception to see what it reveals about realistic goal-setting.
4. **Rationalise the product portfolio**. Five of sixty products generate half of revenue. The long tail contributes little and may be unprofitable to maintain.
5. **Grow Personal Care and SME segments.** They have the best margins in their respective dimensions (−2.99% each). Expanding them improves the overall mix without requiring a price increase.

Evidence supporting the conclusion

- Overall gross margin of ≈ −3.4% across 25,035 transactions (Q1, Q3, Q5, Q8).
- Undiscounted transactions average +568 RWF; at 15–20% discount they average −9,249 RWF (Q6).
- Seven of eight stores achieve ≤80% of target for all 12 months (Q2).
- Electronics accounts for 52.6% of revenue but has the second-worst margin (Q5).
- Personal Care has the best margin (−2.99%) at only 8.1% of revenue (Q5).
- The top five products account for approximately 50% of revenue (from the Pareto analysis in the notebook).

Business interpretation

The data shows a business generating substantial revenue but no positive gross margin on any dimension. This is either a fundamental pricing/cost failure requiring urgent correction, or a data-quality issue requiring correction in the source system. Either way, the correct next action is investigation, not intervention. Once the cost basis is verified, the evidence already in this dataset points clearly to specific areas for margin improvement: Electronics discount policy, Secondary City store economics, and the under-exploited Personal Care and SME segments.

Summary for the CEO

Question	**Key Finding**	**Supporting Evidence**
Q1	Kigali City Centre is the strongest store; Muhanga Hub is the weakest	on Sales, Gross Profit and Achievement (109.4% vs 42.2%).
Q2	Targets are unrealistic for seven of eight stores.	Twelve-month achievement bands at 42–80%.
Q3	Individuals drive volume; SMEs are the most efficient	72.3% of sales vs best margin (−2.99%)
Q4	RFM identifies Champions, Loyal, New, At-Risk and Lost segments.	Recency / Frequency / Monetary quintiles.
Q5	The highest-sales category is not the most profitable.	Electronics 52.6% of sales, −3.38% margin

Q6	Higher discounts are associated with lower profit and no volume uplift.	r = −0.172; banded averages decline.
Q7	Urban channel is more efficient; Secondary City has higher revenue but worse margin.	−3.12% vs −3.71%.
Q8	Institutions are concentrated in Urban stores.	62% of Institutional spend in Urban
Q9	Company-wide negative gross margin.	≈ −3.4% overall.

Conclusions

1. **The company is generating significant revenue but no gross margin on any dimension**. This is the defining finding.
2. **Targets are systematically missed across seven of eight stores.** This is a target-setting problem, not a store-performance problem.
3. **The biggest revenue category (Electronics) is among the least profitable.** The biggest margin lever is here.
4. **Discounting does not drive volume in this data, and is associated with lower profit.** Discount policy warrants review.
5. **Customer, channel, and product-mix distribution is uneven.** SMEs, Personal Care, and potentially Institutions in Secondary Cities are under-exploited.
6. **The next step is investigation, not intervention**. Reconcile `UnitCost` before acting on any finding.